

ORIGINAL ARTICLE

Clinical validation of a tissue-agnostic genome-wide methylome enrichment molecular residual disease assay for head and neck malignancies[☆]

G. Liu^{1,2*}, S. H. Huang¹, L. Ailles^{1,2}, K. Rey-McIntyre¹, C. A. Melton³, S. Y. Shen³, J. M. Burgener³, B. Brown³, J. Zhang³, J. Min³, Y. Wang³, O. Hall³, J. T. Jones³, K. Budhraj³, J. B. Provance³, E. V. Sosa³, A. Licon³, A. Williams³, S. V. Bratman^{1,2}, B. A. Allen³, J. Zhang³, A.-R. Hartman^{3†} and D. D. De Carvalho^{1,2†}

¹Princess Margaret Cancer Centre, University Health Network, Toronto; ²Department of Medical Biophysics, University of Toronto, Toronto, Canada; ³Adela, Inc., Foster City, USA



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Background: Outcomes for patients with locally advanced head and neck cancer (HNC) treated with curative intent remain disappointing, with 5-year survival rates at 50%. Most recurrences occur within the first 2 years after treatment, providing a window of opportunity to identify patients with molecular residual disease (MRD). A tissue-agnostic test for MRD detection in patients with human papillomavirus (HPV) positive and negative HNC, where tissue is often scarce, is needed.

Patients and methods: Patients with stage I-IVB HNC, including patients positive and negative for HPV, were enrolled and peripheral blood plasma was collected longitudinally at diagnosis and ~3, 12, and 24 months after curative intent treatment. The full cohort includes 325 patients with 1155 samples. Samples were split into distinct sets to train and validate a classifier capable of identifying MRD using a tissue-agnostic genome-wide methylome enrichment platform. The primary endpoint was recurrence-free survival (RFS).

Results: With a median follow-up of 60 months, patients in the blinded validation set with MRD positivity experienced significantly worse RFS with a hazard ratio (HR) of 35.7 [95% confidence interval (CI) 10.8-117.8; $P < 0.0001$]. For patients with HPV negativity, HR was 42.3 (95% CI 9.8-182.3; $P < 0.0001$); for patients with HPV-positive oropharyngeal cancer, HR was 24.1 (95% CI 3.0-196.8; $P < 0.0001$). Moreover, the lead time between MRD positivity and clinical recurrence was up to 14.9 months, with a mean lead time of 4.1 months. Surveillance sensitivity was 91% (95% CI 77% to 97%) and specificity was 88% (95% CI 80% to 93%).

Conclusions: Here we validate the clinical performance characteristics of a tissue-agnostic genome-wide methylome enrichment assay for MRD detection in patients with HNC. The MRD detection test showed high sensitivity for identifying recurrence at high specificity across different anatomical sites, HPV status, and treatment regimens, highlighting the broad applicability for MRD detection in patients with HNC.

Key words: cfMeDIP-seq, molecular residual disease, head and neck cancer, cfDNA, precision medicine, liquid biopsy

INTRODUCTION

Head and neck cancers (HNCs) have high relapse rates, with 80%-90% of all recurrences occurring within the first 2 years after the completion of curative-intent treatment.^{1,2} There is an unmet need for biomarkers that can stratify patients with HNC into those who are likely or unlikely to relapse

after definitive curative-intent treatment. Novel approaches to detect relapses before clinical presentation could contribute to improved outcomes by guiding early intervention approaches for relapse. This quest for biomarkers to detect relapses early is complicated by the fact that HNCs are molecularly and etiologically heterogeneous tumors that are commonly divided into cancers driven by human papillomavirus (HPV) and those that are more driven by chronic smoking and alcohol exposure.³ The incidence of HNC is increasing, particularly in younger populations,⁴ further highlighting this unmet need.

Circulating tumor DNA (ctDNA) has emerged as a biomarker to detect and profile residual tumor burden at multiple timepoints throughout the cancer care continuum.⁵

The term molecular residual disease (MRD) has been used to describe the detection of ctDNA and other

*Correspondence to: Dr Geoffrey Liu, Division of Medical Oncology and Hematology, Princess Margaret Cancer Centre, University Health Network, 200 Elizabeth Street, Toronto, Ontario M5G 2C4 Canada. Tel: +1-416-946-4501
E-mail: Geoffrey.Liu@uhn.ca (G. Liu).

†These authors jointly supervised this work.

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biomarkers shed from cancer cells that remain in the body during or after treatment.⁶ MRD detection may help personalize treatments, determine response to treatment, and detect relapse earlier, all of which can lead to improved clinical outcomes.⁷ The first generation of MRD monitoring tests are based on the detection of genomic alterations in tumor tissue, which are then tracked through plasma ctDNA after treatment. Blood-only, tissue-agnostic approaches to MRD monitoring can be implemented more efficiently than tissue-based protocols and do not require access to original tumor tissue or new biopsy procedures. Many HNCs are not surgically resected, thereby limiting the ability to access tissue for tumor-informed approaches.⁸ Tissue-agnostic MRD approaches are ideal for routine clinical use.^{9,10} One currently available tissue-agnostic approach is the detection of HPV-derived DNA as a surrogate for ctDNA in patients with HPV-positive oropharyngeal cancer (OPC).^{11,12} However, this approach is only suitable for the subgroup of patients with HPV-positive tumors, and because it relies only on HPV DNA, it has limited sensitivity. Therefore a tissue-agnostic approach that can detect MRD-positive cases independent of HPV status is needed.

Emerging evidence suggests that cell-free DNA (cfDNA) methylomes provide a biomarker that can fulfill this clinical need. Large studies have demonstrated that the cancer methylome is vastly different from normal tissue,¹³ and cfDNA methylation can detect the presence of ctDNA with high sensitivity without the need for tissue samples or bespoke panel design.^{14–18} One approach to profile the cfDNA methylome is cell-free methylated DNA immunoprecipitation and high-throughput sequencing (cfMeDIP-seq), which uses a tissue-agnostic, genome-wide, methylome enrichment platform. As cfMeDIP-seq does not use chemical or enzymatic conversion, it has multiple advantages over other DNA methylation assays. First, it is nondegrading, preserving more ctDNA signal and cfDNA fragment length information. Second, all four base pairs are sequenced instead of only three, increasing the mapping efficiency and preserving the ctDNA signal. Finally, by sequencing only methylated cfDNA fragments, it avoids wasting sequencing reads at vast regions of the genome without DNA methylation signal, increasing signal to noise.^{15,16} We and others have previously demonstrated the utility of cfMeDIP-seq as a tissue-agnostic approach to detect ctDNA in multiple cancer types and cfDNA sources.^{15–22} In addition, because DNA methylation patterns are intrinsically related to cellular differentiation and cell states, cfMeDIP-seq can also capture molecular patterns reflective of tumor phenotype, behavior, and treatment response.^{9,23,24} Moreover, in a small proof-of-principle study, cfMeDIP-seq demonstrated its potential to longitudinally track the presence of ctDNA in patients with HPV-negative HNC in a tissue-agnostic manner.⁹ However, a blinded clinical validation of this platform for ctDNA detection was previously lacking.

Here, we provide the first blinded clinical validation of this technology. We utilized samples from an existing observational study with longitudinal collection of cfDNA from patients diagnosed with HNC (both HPV-associated and not HPV-associated) who had undergone curative-

intent treatment. cfDNA methylome analysis was conducted to train and validate a classifier for MRD status after treatment. MRD status was then tested for association with clinical outcomes.

PATIENTS AND METHODS

Study design, patients, and samples

This retrospective analysis using banked samples from individuals diagnosed with stage I-IVB HNC was approved by the Research Ethics Board at University Health Network. All patients provided written informed consent.

Patients were recruited from multidisciplinary oncology clinics at Princess Margaret Cancer Centre in Toronto, Canada, from 2014 to 2019. Eligible patients included those >18 years of age with stage I-IVB HNC, according to Union for International Cancer Control/American Joint Committee on Cancer (UICC/AJCC) eighth edition staging criteria.²⁵ HPV status was determined from OPC tumor tissue biopsies using p16 immunohistochemical assay as previously described.²⁶ HPV status for other anatomic sites was not assessed, per standard clinical care. Patients diagnosed with Epstein–Barr virus-associated nasopharyngeal carcinoma were excluded.

All patients received curative-intent treatment consisting of either surgery, radiation therapy with or without chemotherapy, or surgery followed by adjuvant radiation therapy with or without chemotherapy as shown in [Supplementary Figure S1](https://doi.org/10.1016/j.annonc.2024.08.2348), available at <https://doi.org/10.1016/j.annonc.2024.08.2348>. Patients with distant metastatic disease at the time of diagnosis and those treated with palliative intent were excluded. All patients had a blood draw at baseline (i.e. before curative intent therapy), and mean 3.0 months (range 0.7–8.5 months) after completion of therapy. A portion of the patients had additional blood draws ~12 and 24 months after therapy.

Patient treatment paradigms reflecting real-world practice are shown in [Supplementary Figure S1](https://doi.org/10.1016/j.annonc.2024.08.2348), available at <https://doi.org/10.1016/j.annonc.2024.08.2348>. Treatment decisions were at the discretion of the treating physician and were not proscribed by the study protocol. Recurrence status was determined as part of standard-of-care clinical practice for all patients. Recurrences were typically established based on histologic evidence of recurrence (biopsy proven). When a biopsy was not feasible, unequivocal radiologic or clinical evidence was used to confirm recurrence.

Patients were randomized to be placed into one of two groups: a training dataset, designed to develop a biomarker classifier for HNC MRD, and an independent, hold-out validation dataset, in approximately equal patient numbers ([Figure 1](https://doi.org/10.1016/j.annonc.2024.08.2348)).

Sample processing and assay details

Peripheral blood plasma was isolated from ethylenediamine tetraacetic acid tubes within 2 h of collection and stored at –80°C until further processing. cfDNA was isolated from

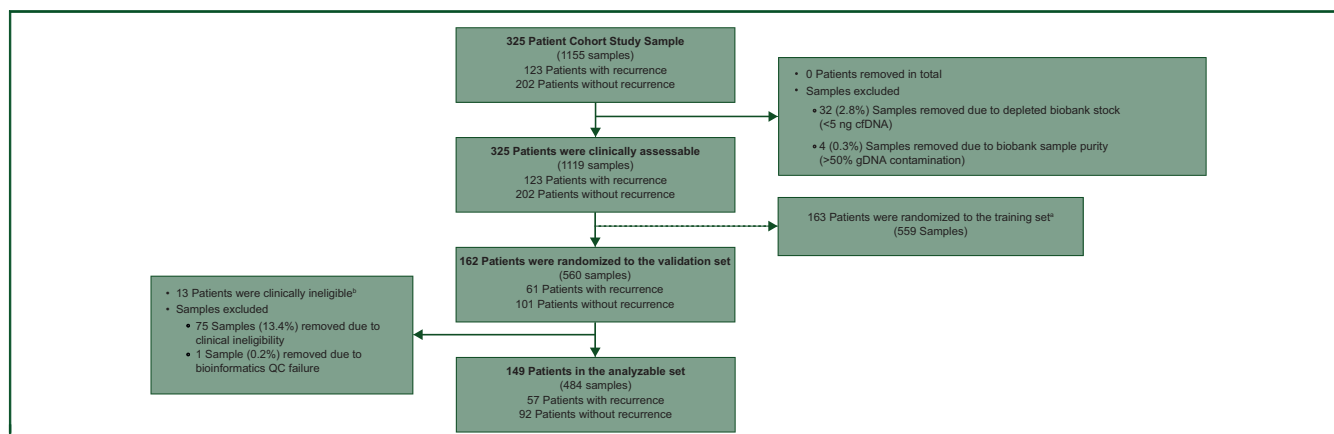


Figure 1. Patients with head and neck cancer and their banked plasma samples.

cfDNA, cell-free DNA; gDNA, genomic DNA; QC, quality control.

^aSee [Supplementary Materials](https://doi.org/10.1016/j.annonc.2024.08.2348), available at <https://doi.org/10.1016/j.annonc.2024.08.2348> for additional information on the training set and definitions of each cohort.

^bSee [Supplementary Table S1](https://doi.org/10.1016/j.annonc.2024.08.2348), available at <https://doi.org/10.1016/j.annonc.2024.08.2348>, for full details of ineligible patients.

plasma, and 5–10 ng of cfDNA was used for methylation analysis using a bisulfite-free, nondegradative genome-wide DNA methylation enrichment platform based on cfMeDIP-seq ([Supplementary Figure S2](https://doi.org/10.1016/j.annonc.2024.08.2348), available at <https://doi.org/10.1016/j.annonc.2024.08.2348>). Sequencing was completed by the Princess Margaret Genomics Centre, Toronto, Canada (www.pmggenomics.ca).

Using the training dataset, we then developed and refined a classifier ([Figure 1](https://doi.org/10.1016/j.annonc.2024.08.2348)) that utilizes blood samples collected before and after treatment from patients with confirmed recurrence and nonrecurrence; the classifier was trained by incorporating regions of interest that maximized ctDNA-derived MRD positive signal. To identify robust regions of interest, we carried out repeated sixfold cross-validation utilizing cfMeDIP-seq profiles from baseline plasma samples ($n = 135$) and post-treatment plasma samples of patients with clinical recurrence ($n = 78$) as cases, and cfMeDIP-seq profiles from post-treatment plasma of patients with no clinical recurrence ($n = 246$) as controls, from the training set. See [Supplementary Appendix](https://doi.org/10.1016/j.annonc.2024.08.2348), available at <https://doi.org/10.1016/j.annonc.2024.08.2348> for detailed information.

The algorithm producing MRD Detection Scores was selected and locked based on the training data ([Figure 1](https://doi.org/10.1016/j.annonc.2024.08.2348)). As prespecified in the statistical analysis plan (SAP), the cut-off was selected to allow for a 5% false positive rate at the first postcurative intent (mean 3 months) timepoint. The cut-off selected maximizes the sensitivity with the constraint of an acceptable specificity. Test performance of the locked classifier was assessed in an independent holdout sample set blinded to the laboratory and the bioinformatics departments ([Figure 1](https://doi.org/10.1016/j.annonc.2024.08.2348)).

Endpoints

The primary endpoint for this study is the difference in recurrence-free survival (RFS) by MRD test result status through surveillance monitoring. RFS was measured from

the date of treatment completion to the date of recurrence or death due to HNC, whichever came earlier, and was censored at last follow-up or non-HNC-related death. Overall survival (OS) and patterns of failure (locoregional versus distant) were exploratory endpoints.

MRD detection status during surveillance monitoring was defined as previously described.²⁷ Patients had MRD detected if they had at least one follow-up timepoint with an MRD-positive result. A patient was considered not to have MRD detected only if all blood draws through multiple follow-up timepoints had MRD negative results. Patients with a single post-treatment follow-up timepoint that were MRD not detected were excluded from the surveillance analysis.²⁷

Statistical methods

A prespecified SAP was signed off before the analysis of the validation data. Descriptive statistics were reported for demographics and baseline characteristics. For continuous variables, minimum, maximum, median, and interquartile ranges were presented; for categorical variables, the count and percentage of patients in each category were calculated. Each sample was assigned a label of MRD detected or MRD not detected by comparing its MRD detection score with a cut-off value. The cut-off was chosen to allow for a 5% false positive rate at the first postcurative intent treatment timepoint (mean 3 months), which was prespecified in the SAP. This cut-off strikes a balance between sensitivity and specificity, ensuring adequate clinical utility.

The Kaplan–Meier method was used to compare the RFS between the MRD detected and MRD not detected groups, and the two-sided log-rank test was used to assess statistical significance. Differences were considered statistically significant at a $P < 0.05$. The hazard ratio (HR) between the two groups was estimated using the Cox proportional hazards model. OS was evaluated in a similar manner.

Multivariate analysis for RFS was conducted by adding statistically significant clinical variables obtained from univariate analyses. The likelihood ratio test was performed using nested models with and without MRD detection. Any missing values were included in the category ‘missing’ and were neither imputed nor excluded. The proportional hazards assumption was tested with a global test of Schoenfeld residuals.

Performance metrics included sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV). Their 95% confidence intervals (CIs) were calculated using the Wilson (score) method. Fisher’s exact test was conducted to test whether any performance metric was significantly different across subpopulations. The Breslow–Day test was used to assess if the association between recurrence status and MRD detection was homogeneous across subpopulations. Maximum and mean lead times were summarized. Statistical analyses were conducted using R statistical software (R Foundation, Vienna, Austria), version 4.2.2.

RESULTS

Patients

A total of 325 patients were included in the study. The training set included 163 patients, and the validation set 162 patients (Figure 1). Additional details for patients who were clinically ineligible are given in [Supplementary Table S1](#), available at <https://doi.org/10.1016/j.annonc.2024.08.2348>.

In the analyzable dataset ($n = 149$), the majority of patients were 50–69 years of age (66%), White (71%), and male (77%). Of those who reported smoking history, 68% were former or current smokers (Table 1). Early-stage patients were over-represented (54% stage I–II). The cohort represented various anatomic subtypes, with oropharyngeal (48%) and lip/oral cavity (23%) cancers being the most common. As expected, the majority of OPCs were HPV-positive (75%). Most patients (95%) had squamous cell carcinoma. Overall, the sample failure rate due to quality control was low (0.2% or 1/560).

The mean time to first post-treatment blood draw was 2.9 months (range 0.9–7.8 months); 146 of 149 patients had analyzable first post-treatment blood draw while 109 patients had a blood draw at ~12 months and 89 at ~24 months. The median follow-up for the analyzable set was 60.0 months. In patients with a recurrence event or death due to HNC, the median follow-up was 17.0 months. The remaining patients had a median follow-up time of 71.2 months.

ctDNA analysis

Using the unblinded training set ($n = 163$ patients), a classifier was developed to detect the presence of ctDNA methylome signal in pretreatment samples and post-treatment recurrence samples (MRD positive) as compared with post-treatment nonrecurrence samples (MRD negative)

Table 1. Patient demographics and clinical characteristics at time of diagnosis

Variable		Validation dataset
Total, N		149
Age at diagnosis (years)	Median (Q1–Q3)	62 (53–69)
	Range	26–88
Sex, n (%)	Female	34 (22.8)
	Male	115 (77.2)
Stage (AJCC eight edition), n (%)	I	50 (33.6)
	II	31 (20.8)
	III	24 (16.1)
	IVA–IVB	44 (29.5)
Treatment, n (%)	CRT	51 (34.2)
	RT	49 (32.9)
	Surgery	19 (12.8)
	Surgery + CRT	11 (7.4)
	Surgery + RT	19 (12.8)
Cancer site/HPV status, n (%)	Hypopharynx	6 (4.0)
	Larynx	21 (14.1)
	Lip and oral cavity	35 (23.5)
	Nasal cavity/paranasal sinus	6 (4.0)
	Nasopharynx	1 (0.7)
	Oropharynx:	72 (48.3)
	HPV positive	54 (75.0)
	HPV negative	17 (23.6)
	No data	1 (1.4)
	Salivary glands	4 (2.7)
	Unknown primary	4 (2.7)
Histology, n (%)	Squamous cell carcinoma	141 (94.6)
	Nonsquamous cell carcinoma	8 (5.4)
Race, n (%)	White	106 (71.1)
	Non-White	22 (14.8)
	Missing	21 (14.1)
Smoking status, n (%)	Current	27 (18.1)
	Former	58 (38.9)
	Never	40 (26.8)
	Missing	24 (16.1)
Alcohol status, n (%)	Current	68 (45.6)
	Former	31 (20.8)
	Never	27 (18.1)
	Missing	23 (15.4)

Training and validation set definitions and details are presented under the Methods section and in the [Supplementary Materials](#), available at <https://doi.org/10.1016/j.annonc.2024.08.2348>.

AJCC, American Joint Committee on Cancer; CRT, chemoradiotherapy; HPV, human papillomavirus; Q1, first quartile; Q3, third quartile; RT, radiotherapy.

as described in [Supplementary Appendix](#), available at <https://doi.org/10.1016/j.annonc.2024.08.2348>. Furthermore, the differentially methylated regions identified during this classifier development were hypermethylated in HNC tissue ($n = 523$ patients) compared with adjacent normal tissue ($n = 45$ patients) and compared with healthy peripheral blood leucocytes ($n = 93$ noncancer controls; $P = < 0.0001$) in an external independent dataset from the Cancer Genome Atlas ([Supplementary Figure S3](#), available at <https://doi.org/10.1016/j.annonc.2024.08.2348>), indicating the classifier is detecting the presence of ctDNA methylome signal.

Association of MRD detection and clinical outcomes

ctDNA analysis was carried out on post-treatment samples throughout surveillance to assess whether MRD detection status was associated with clinical outcomes. Patients with

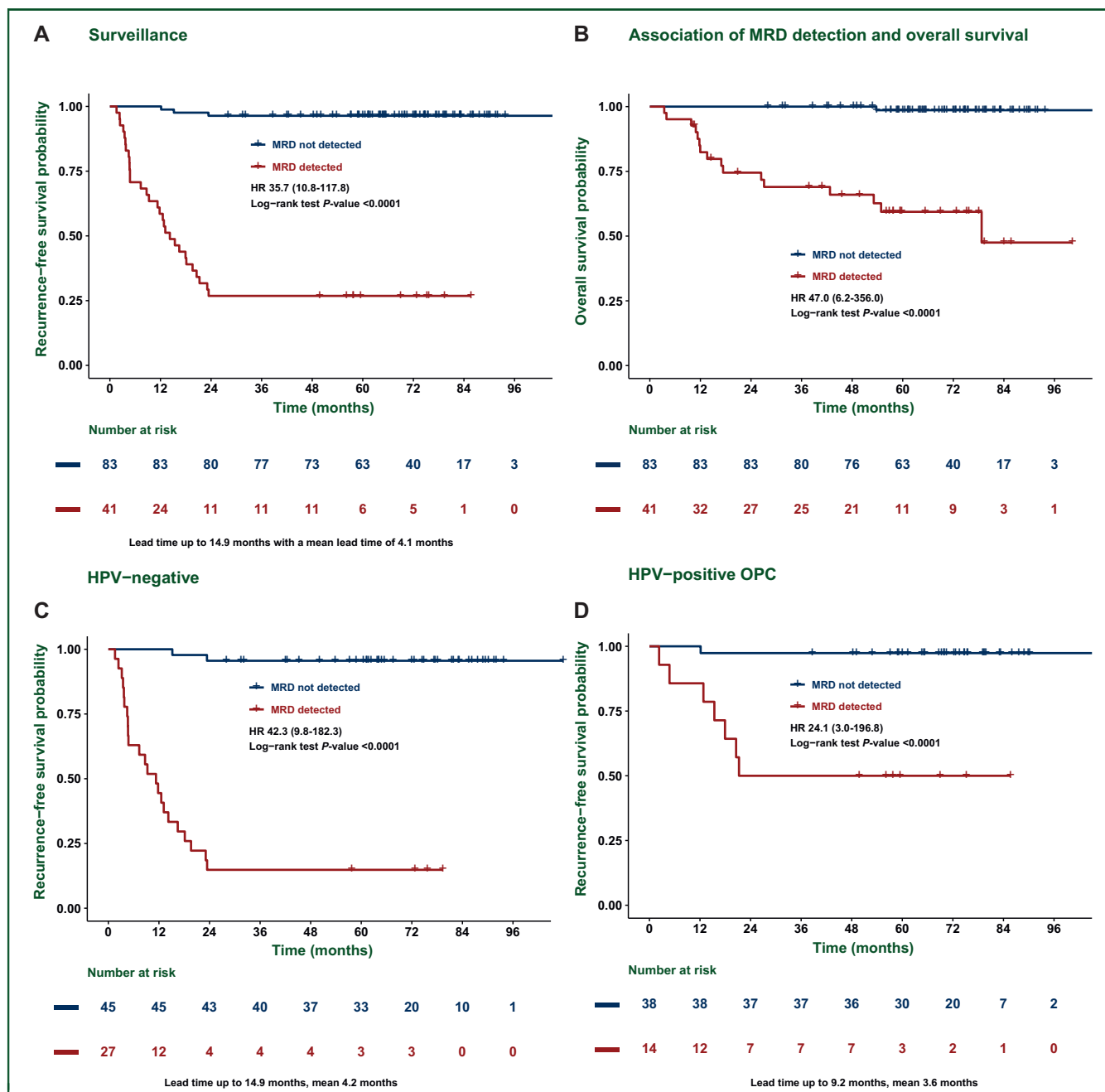


Figure 2. Kaplan–Meier estimates of time to event by MRD surveillance detection in the validation set. (A) RFS in all patients ($n = 124$), (B) OS in all patients ($n = 124$), (C) HPV-negative subgroup ($n = 72$), and (D) HPV-positive subgroup ($n = 52$). HPV-negative subgroup included HPV-negative OPC plus other head and neck cancer sites. HR followed in parentheses by 95% confidence intervals.

HPV, human papillomavirus; HR, hazard ratio; MRD, molecular residual disease; OPC, oropharyngeal cancer; OS, overall survival; RFS, recurrence-free survival.

MRD detected had a significantly reduced RFS compared with those where MRD was not detected (HR 35.7, 95% CI 10.8-117.8; log-rank $P < 0.0001$; Figure 2A). We observed a similarly strong association between MRD detection and OS (HR 47.0, 95% CI 6.2-356.0; log-rank $P < 0.0001$; Figure 2B). In a subset analysis, significantly worse RFS was consistently observed in patients who were MRD detected despite HPV status: for patients with HPV-negative disease, HR was 42.3 (95% CI 9.8-182.3; log-rank $P < 0.0001$); for patients with HPV-positive OPC, HR was 24.1 (95% CI 3.0-196.8; log-rank $P < 0.0001$; Figure 2C and D). Lead time from a positive MRD

result and an RFS event was up to 14.9 months with a mean of 4.1 months.

Performance

The performance metrics are summarized in Table 2. In the overall population, surveillance sensitivity was 91% (95% CI 77% to 97%), specificity was 88% (95% CI 80% to 93%), PPV was 74% (95% CI 59% to 85%), and NPV was 96% (95% CI 90% to 99%). No heterogeneous association between recurrence status and MRD status across HPV strata and

Table 2. Performance of MRD detection by cfMeDIP-seq assay overall, and by HPV status and treatment subsets

	Specificity (95% CI), %	Sensitivity (95% CI), %	PPV (95% CI), %	NPV (95% CI), %	Breslow–Day <i>P</i> value
Overall (<i>n</i> = 125)	88 (80 to 93)	91 (77 to 97)	74 (59 to 85)	96 (90 to 99)	
HPV status					0.39
HPV+ OPC (<i>n</i> = 52)	84 (71 to 92)	88 (53 to 98)	50 (27 to 73)	97 (87 to 100)	
HPV– HNC (<i>n</i> = 73)	91 (80 to 97)	92 (76 to 98)	86 (69 to 94)	96 (85 to 99)	
Fisher's exact test <i>P</i> value	0.34	>0.99	0.02	>0.99	
Treatment					0.73
CRT (<i>n</i> = 42)	94 (80 to 98)	89 (57 to 98)	80 (49 to 94)	97 (84 to 99)	
RT (<i>n</i> = 41)	81 (65 to 91)	89 (57 to 98)	57 (33 to 79)	96 (82 to 99)	
Surgery (<i>n</i> = 18)	82 (52 to 95)	100 (65 to 100)	78 (45 to 94)	100 (70 to 100)	
Surgery + CRT (<i>n</i> = 8)	100 (57 to 100)	100 (44 to 100)	100 (44 to 100)	100 (57 to 100)	
Surgery + RT (<i>n</i> = 16)	90 (60 to 98)	83 (44 to 97)	83 (44 to 97)	90 (60 to 98)	
Fisher's exact test <i>P</i> value	0.48	0.91	0.60	0.73	

cfMeDIP-seq, cell-free methylated DNA immunoprecipitation and high-throughput sequencing; CI, confidence interval; CRT, chemoradiotherapy; HNC, head and neck cancer; HPV, human papillomavirus; MRD, molecular residual disease; NPV, negative predictive value; OPC, oropharyngeal cancer; PPV, positive predictive value; RT, radiotherapy.

treatment regimens strata was found based on the Breslow–Day test ($P = 0.39$ and 0.73 , respectively). Regarding specific performance across strata, sensitivity and specificity were not significantly different across HPV status ($P > 0.99$ and 0.34) and treatment regimens ($P = 0.91$ and 0.48), according to Fisher's exact test. NPV was not significantly different across subpopulations either ($P > 0.99$ and 0.73). PPV was not significantly different across treatment regimens ($P = 0.60$) but was higher in patients with HPV-negative disease. Sensitivity by primary location of recurrence (distant versus locoregional) site is shown in [Supplementary Table S2](https://doi.org/10.1016/j.annonc.2024.08.2348), available at <https://doi.org/10.1016/j.annonc.2024.08.2348>.

Univariate and multivariate analyses

The univariate analysis shown in [Table 3](#) included known demographic and clinical risk factors for HNC recurrence. In

addition to MRD detection, stage, HPV status, race, age, and smoking history were significantly associated with RFS in the univariate analysis in this patient population. In the multivariate analysis, which included all significant factors identified in the univariate analysis, the highest predictive value was observed for MRD detection ($P < 0.0001$). The likelihood ratio test based on nested models with and without MRD detection status yielded a $P < 0.0001$, demonstrating the highly significant benefit to adding MRD detection status to the prediction of recurrence after accounting for clinical factors.

Changes in MRD detection score correlate with outcomes

[Figure 3A](#) shows the trajectory of the MRD detection score from pre-treatment (T0) to post-treatment (T1–T3) samples. High MRD detection score was observed at T0 for patients with and without recurrences. For patients who did not

Table 3. Univariate and multivariate Cox proportional hazard models

Variable	Univariate hazard ratio (95% CI)	Univariate <i>P</i> value	Multivariate hazard ratio (95% CI)	Multivariate <i>P</i> value
MRD detected (<i>n</i> = 41) versus not detected (<i>n</i> = 83)	35.7 (10.8–117.8)	<0.0001	34.6 (9.6–124.5)	<0.0001
Stage III–IVB (<i>n</i> = 48) versus stage I–II (<i>n</i> = 76)	5.6 (2.6–12.0)	<0.0001	4.1 (1.5–11.2)	0.006
HPV+ oropharynx (<i>n</i> = 52) versus HPV– (<i>n</i> = 72)	0.4 (0.2–0.9)	0.02	0.7 (0.3–1.7)	0.38
Non-White (<i>n</i> = 19) versus White (<i>n</i> = 86)	2.3 (0.9–5.7)	0.07	1.0 (0.4–2.7) ^a	0.97
Missing (<i>n</i> = 19) versus White (<i>n</i> = 86)	4.4 (2.0–9.7)	0.0002	1.3 (0.3–5.8)	0.72
Age (<i>n</i> = 124) per year increase	1.0 (1.0–1.1) ^b	0.007	1.0 (1.0–1.0) ^c	0.82
Ever-smoker (<i>n</i> = 69) versus never-smoker (<i>n</i> = 36)	1.2 (0.5–2.9)	0.71	1.3 (0.5–3.6)	0.63
Missing (<i>n</i> = 19) versus never-smoker (<i>n</i> = 36)	3.4 (1.3–9.0)	0.01	1.6 (0.3–9.3)	0.59
N1 (<i>n</i> = 44) versus N0 (<i>n</i> = 44)	0.6 (0.2–1.5)	0.26		
N2 (<i>n</i> = 31) versus N0 (<i>n</i> = 44)	1.7 (0.7–3.8)	0.21		
N3 (<i>n</i> = 5) versus N0 (<i>n</i> = 44)	3.4 (1.0–12.4)	0.06		
Female (<i>n</i> = 29) versus male (<i>n</i> = 95)	0.7 (0.3–1.7)	0.43		
Lip and oral cavity (<i>n</i> = 29) versus other primary site (<i>n</i> = 33)	0.8 (0.3–1.9)	0.56		
Oropharynx (<i>n</i> = 62) versus other primary site (<i>n</i> = 33)	0.5 (0.2–1.2)	0.12		
Surgery (<i>n</i> = 18) versus C/RT (<i>n</i> = 82)	2.2 (0.9–5.3)	0.08		
Surgery + adjuvant C/RT (<i>n</i> = 24) versus C/RT (<i>n</i> = 82)	2.0 (0.9–4.5)	0.09		

HRs are presented with 95% CI. For univariate analysis, each row is a single model. For multivariate analysis, the entire column is a single model. Ever smoker: included current or former smokers, an adult who has or had smoked in his or her lifetime (more than 100 cigarettes in their lifetime). Included cigarettes or cigars. never smoker: never smoked (did not smoke more than 100 cigarettes in their lifetime).

C/RT, radiation with or without chemotherapy; CI, confidence interval; HPV+, human papillomavirus positive; HR, hazard ratio; MRD, molecular residual disease; N0, N1, N2, and N3, nodal staging.

^aHR (95% CI) in two decimals: 0.98 (0.36–2.67).

^bHR (95% CI) in two decimals: 1.04 (1.01–1.08).

^cHR (95% CI) in two decimals: 1.00 (0.96–1.04).

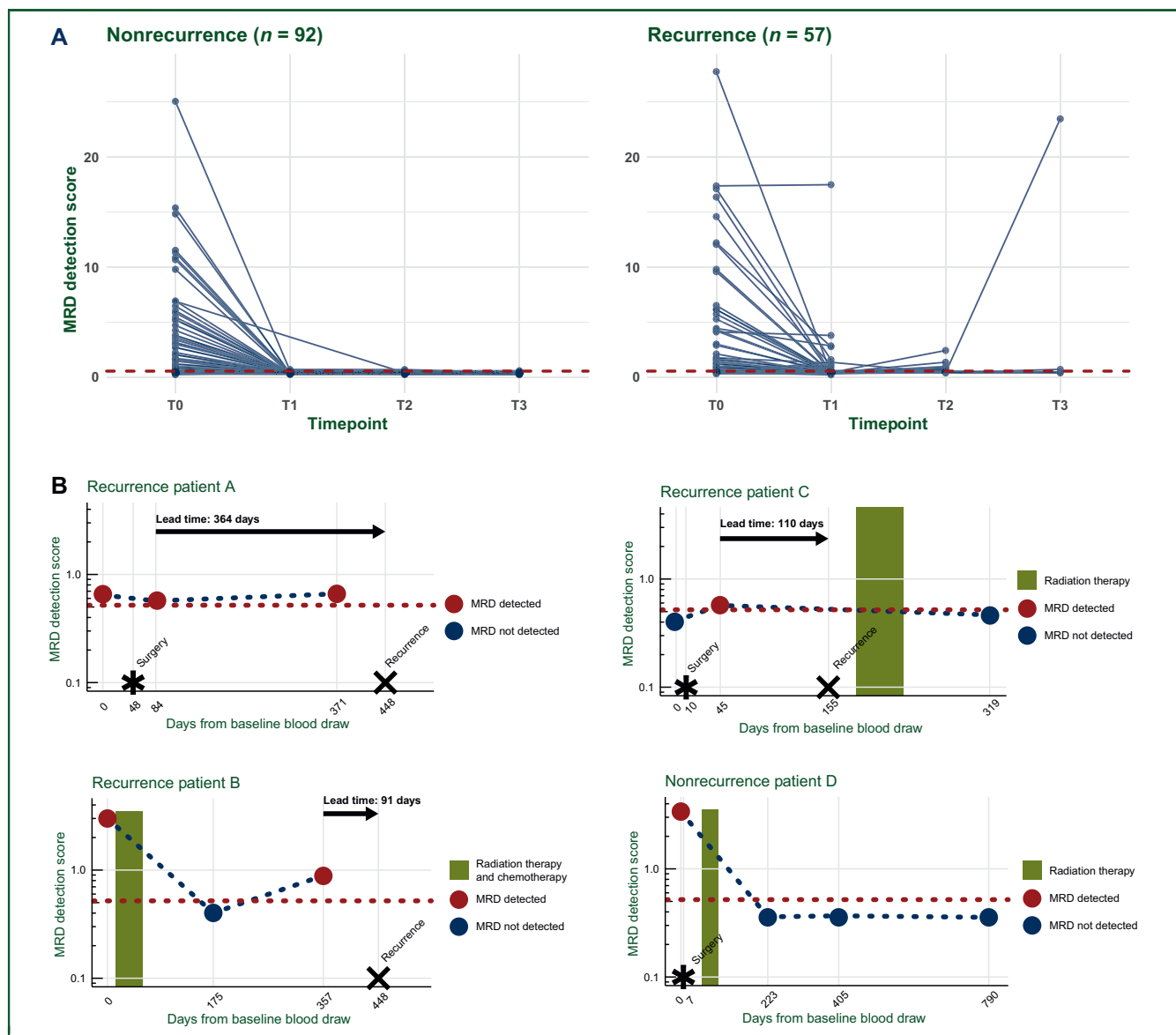


Figure 3. Changes in MRD detection score correlate with outcomes. (A) The trajectories in MRD detection scores are displayed for nonrecurring and recurring patients. The red dashed line showed the threshold to determine whether a sample had MRD detected or not detected. (B) Representative case studies: MRD detection in four patients. All four patients had squamous cell carcinomas. Patient A had stage III lip and oral cavity cancer, HPV status unknown. Patient B had stage III HPV-positive oropharyngeal cancer. Patient C had stage I HPV-positive oropharyngeal cancer. Patient D had stage IVA lip and oral cavity cancer with no recurrence. The red dashed line shows the threshold to determine whether a sample had MRD detected or not detected. The star represents the date of surgery. A shaded rectangular area represents the period for radiation or chemoradiation. Dots represent MRD detection scores at specific numbers of days from baseline pretreatment blood draw; the red dot represents MRD detected; the blue dot represents MRD not being detected. An X marks the day of recurrence. Lead times are also presented on the graphs with recurrences. HPV, human papillomavirus; MRD, molecular residual disease.

recur, MRD detection score quickly dropped after the completion of treatment and remained low across surveillance. For patients who recurred, MRD detection score also dropped for many patients after treatment (with some remaining above the threshold). Most patients who recurred had an increase in their MRD detection score during surveillance. There were 15 recurring patients who were negative at T1 and were then correctly detected at T2 or T3.

To illustrate the potential utility of the MRD test in the context of distinct clinical scenarios, we plotted the trajectories of MRD detection scores in three patients who experienced recurrence and one patient who remained

recurrence free (Figure 3B). The red dashed line represents the threshold to determine whether a sample had MRD detected or not detected. MRD detection score for recurrent patient A was high before treatment, dropped after surgery, and increased before recurrence. All three blood samples had scores higher than the threshold. MRD was detected 364 days ahead of confirmed recurrence. Patient B also showed a V shape trajectory but the first blood draw after chemoradiotherapy had an MRD score below the threshold. MRD was detected 91 days ahead of confirmed recurrence. The MRD score was below the threshold for patient C with stage I HPV+ OPC, before surgery but

increased after surgery, 110 days ahead of confirmed recurrence. The MRD detection score fell below the threshold after radiation treatment for recurrence. Patient D had detectable MRD before surgery but undetectable levels throughout the follow-up and did not experience a recurrence.

DISCUSSION

Outcomes for patients with locally advanced HNC remain disappointing, with 5-year OS rates at 50%.^{1,28,29} Curative treatment involves unimodality or multimodality treatments involving surgery, radiation, and/or chemotherapy. Moreover, ~80%-90% of all recurrences occur within the first 2 years after completion of treatment, creating a unique opportunity to identify residual disease during this time window. Personalized treatment decisions based on MRD detection could improve outcomes.

We report the most extensive blinded clinical validation analysis of MRD in patients with HNC, inclusive of both HPV-negative and HPV-positive cancers. MRD was detected using a tissue-agnostic genome-wide methylome enrichment platform from plasma cfDNA. We demonstrate that MRD positivity using this platform predicts and detects recurrence and provides relative quantification across HNCs, including both HPV-negative cancer sites and HPV-positive subgroups of patients.

This study represents a real-world cohort with prospectively collected samples at a high-volume cancer center. This study enrolled 325 patients who underwent treatment paradigms representing a diverse group of HNCs with clinical follow-up of sufficient duration to assess the impact of MRD status on disease recurrence.

The approach we present here differs from other available tests for MRD detection in HNCs in multiple aspects. First, it does not require tumor tissue to create a bespoke panel for MRD testing. Second, it requires low-input starting material consisting of 10 ng of cfDNA, which can be obtained readily from plasma derived from one 10-mL blood collection tube. Using this approach, laboratory failure rates were extremely low (<0.4%) and primarily due to genomic DNA contamination from sample collection, and therefore did not adversely impact analysis in this study. The expected failure rate should be even lower with prospective sample collection and real-time MRD testing. This failure rate is dramatically lower than what has been observed in tumor-informed approaches where up to 20% of samples or one in five patients are not eligible for the test.^{1,30} This is especially relevant in HNCs where a significant portion of diagnoses are made through small biopsy samples and are not surgically resected. Moreover, as curative intent treatment for HNCs moves to the neoadjuvant setting,³¹ a test that does not rely on tumor tissue and bespoke panels will be easier to implement. In sum, our study underlines the potential generalizability of a tissue-agnostic genome-wide methylome enrichment platform across the varied diagnostic and treatment approaches observed with HNC.

In this study, we observed that MRD detection is a significant and clinically useful predictor of poor RFS. Significant differences in RFS were observed, with an HR of 35.7 ($P < 0.001$) during surveillance. The lead time between ctDNA positivity and clinical recurrence was up to 14.9 months, with a mean lead time of 4.1 months. The MRD detection test showed robust and high sensitivity and specificity for identifying recurrence (overall 91% sensitivity at 88% specificity) across different anatomical sites, treatment modalities, and HPV status in patients with HNC. This performance is comparable to recent data reported using tissue-informed approaches,¹ where surveillance sensitivity in patients with predominantly late-stage HPV-negative HNC was 88%. In our cohort, 54% of patients had stage I and II HNC, representing both HPV-positive OPC and HPV-negative cases. Importantly, MRD detection remained a significant predictor of RFS after accounting for clinical variables widely accepted to predict outcomes in HNC (e.g. stage, HPV status; $P < 0.0001$).

Sensitivity, specificity, and NPV were not statistically different when stratified by HPV status. PPV was not significantly different across treatment regimens but was higher in patients with HPV-negative cancer, driven mainly by the higher recurrence rates in this subpopulation. The robust sensitivity across all subtypes and treatment regimens suggests the utility of this test to predict relapse ahead of clinical presentation to enable earlier treatment. The lower PPV in patients with HPV-positive OPC would not lead to safety issues as the test would be used to triage imaging to confirm recurrence rather than an escalation of treatment. Significant differences in OS were observed, with an HR of 47.0 ($P < 0.0001$) during surveillance.

Currently, the only clinically available tissue-agnostic MRD test for patients with HNC is a PCR-based test for HPV-positive OPC,^{11,12} which accounts for ~20% of all HNC diagnosed in the United States.³²⁻³⁵ Most new diagnoses of HNCs fall outside this subtype, underscoring the need for new tools for HNC to be applied broadly across the HNC population.

After accounting for clinical variables widely accepted to predict outcomes in HNC (e.g. stage, HPV status), a nested multivariate model demonstrated that the MRD test result significantly improves prediction beyond these clinical variables ($P < 0.0001$).

Limitations of the present study are the following: a single-center study, limited blood draw timepoints, and a retrospective analysis of prospectively collected samples. The limited number of blood draws potentially underestimates the sensitivity of the assay compared with scenarios in which surveillance involves more frequent testing. Future prospective studies will yield additional evidence for the use and clinical utility of this approach.

Altogether, the results presented here validate the clinical performance characteristics of a tissue-agnostic genome-wide methylome enrichment assay for MRD detection in patients with HNC. This approach produced high performance across different sites, HPV status, and treatment

regimens, highlighting the broad applicability for MRD detection in patients with HNC.

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DISCLOSURE

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